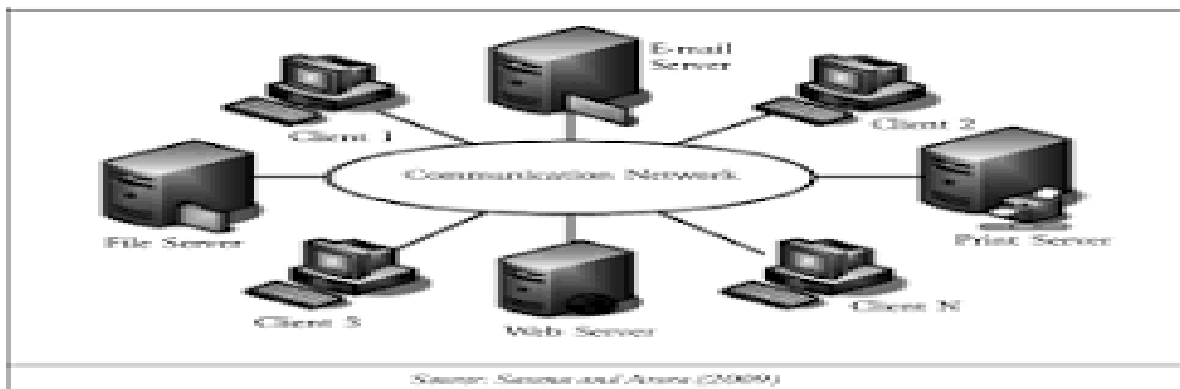


CH-1 INTRODUCTION TO COMPUTER NETWORK & IT'S APPLICATION**Q-1) What is a computer network?**

A computer network is a collection of two or more computer systems that are linked together. A network connection can be established using either cable or wireless media. Hardware and software are used to connect computers and tools in any network.

Each computing device in a network is called a node or station. The nodes can be routers, personal computers, and servers. Data transformation is done via the network using rules known as protocols. The protocols are the set of rules which every node of the network should follow for transferring information over the wired or wireless network.

**How Does a Computer Network Work?**

Computer Networks simply work using nodes and links. Data Communication Equipment is simply termed as Nodes. For example, Modems, Hubs, Switches, etc. whereas links in Computer networks can be referred to as a connection between two nodes. We have several types of links like cable wires, optical fibers, etc.

Whenever a Computer Network is working, nodes have the work of sending and receiving data via the links. Computer Network provides some set of protocols that helps in following the rules and protocols.

Uses of Computer Network:

- It allows you to share resources such as printers, scanners, etc.

- You can share expensive software and database among network users.
- It facilitates communications from one computer to another computer.
- It allows the exchange of data and information among users through a network.

Q-2) Explain Types of computer networks

There are several types of computer networks. Which network an organization uses depends on factors such as the number of devices, types of operating systems, transmission medium used, network topology, the distance between each device and their geographic scale.

Popular Computer Networks:

- Local Area Network (LAN)
- Metropolitan Area Network (MAN)
- Wide Area Network (WAN)

Local Area Network (LAN):

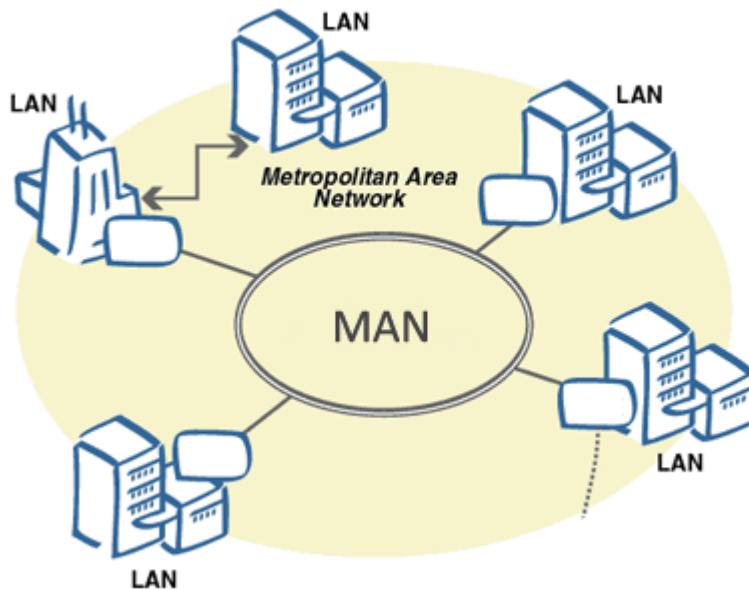
As the name suggests, the local area network is a computer network that operates in a small area, i.e., it connects computers in a small geographical area like within an office, company, school, or any other organization. So, it exists within a specific area, e.g. home network, office network, school network, etc.

A [LAN](#) is a network that covers an area of around 10 kilometers.

A local area network may be a wired or wireless network or a combination of both. The devices in a LAN are generally connected using an Ethernet cable, which offers an interface to connect multiple devices like router, switches, and computers. For example, using a single router, few Ethernet cables, and computers, you can create a LAN at your home, office, etc. In this network, one computer may act as a server and other computers, which are part of the network, may serve as clients.



Metropolitan Area Network (MAN):



MAN is a high-speed network that spreads over a large geographical area such as a metro city or town. It is set up by connecting the local area networks using routers and local telephone exchange lines. It can be operated by a private company, or it may be a service provided by a company such as a local telephone company.

The area covered by MAN is larger than the LAN but smaller than a WAN. Its network ranges from 5 to 50 km.

Examples of MAN:

- Cable TV Network
- Telephone service providers that provide high-speed DSL lines
- Connected fire stations in a city
- Connected branches of a school in a city

Wide Area Network (WAN):



WAN extends over a large geographical area. It is not confined within an office, school, city, or town and is mainly set up by telephone lines, fiber optic, or satellite links. It is mostly used by big organizations like banks and multinational companies to communicate with their branches and customers across the world. Although it is structurally similar to MAN, it is different from MAN in terms of its range, e.g., MAN covers up to 50 Kms, whereas WAM covers distances larger than 50 Km, e.g., 1000km or more.

The internet is considered the largest WAN in the world.

The computers are connected to the wide area network through public networks, such as telephone systems, leased lines or satellites. The users of a WAN do not own the network as it is a large setup connecting the remote computer systems. However, they are required to subscribe to a service provided by a telecommunication provider to use this network.

Examples of WAN:

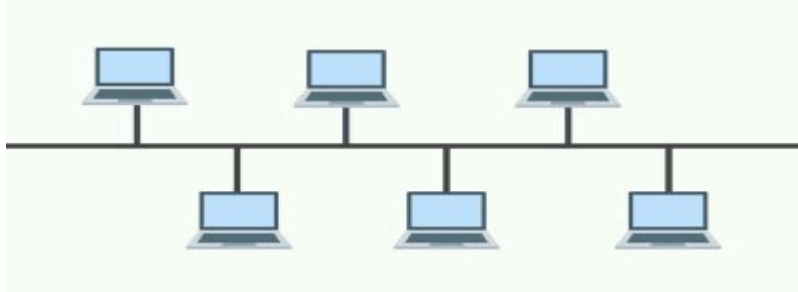
- Internet
- US defense department
- Stock exchanges network
- Railway reservation system
- Big Banks' cash dispensers' network
- Satellite systems

Q-3) Network Topology

Topology: It refers to the physical arrangement and representation of all the nodes and components of the network. In general terms, Topology defines the structure of the entire network. The network topology is

divided into five types. They are bus topology, star topology, ring topology, mesh topology, and tree topology.

1) Bus Topology:



In this arrangement, the nodes (computers) are connected through interface connectors to a single communication line (central cable) that carries the message in both the directions. The central cable to which all the nodes are connected is the backbone of the network. It is called a bus. The signal in this arrangement travels in both directions to all the machines until it finds the recipient machine. It is easy to set up than other topologies as it uses only a single central cable to establish the network.

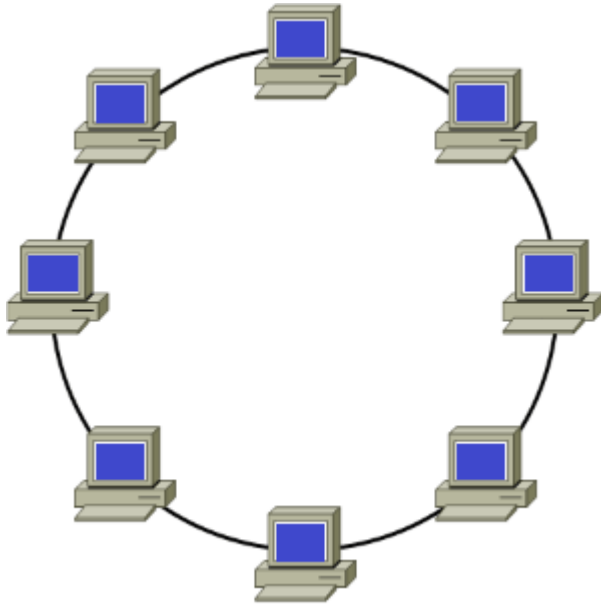
Advantages:

- Configuration of the network is easy.
- Less costly because a single cable is used to connect all nodes.
- The Bus topology supports a maximum of 10 Mbps speed by using the network's coaxial or twisted pair cables.

Disadvantages

- Due to the multipoint communication model, it is difficult to identify and isolate the faulty terminals.
- The reconfiguration will affect the network and slows down the performance.
- Signal interference is another drawback of the bus topology; if two or more nodes transmit the messages simultaneously, their signals will collide.
- A single node failure also causes the breakdown of the whole network.

2) Ring Topology:



As the name suggests, in a ring topology, the computers are connected in a circular and closed loop. The message in this topology moves only in one direction around the ring from one node to another node and is checked by each node for a matching destination address. So, the data keeps moving until it reaches its destination. All nodes are equal; a client-server relationship does not exist between them. As the nodes are in the form of a ring, if one node fails to transmit the data, the flow of communication is severed.

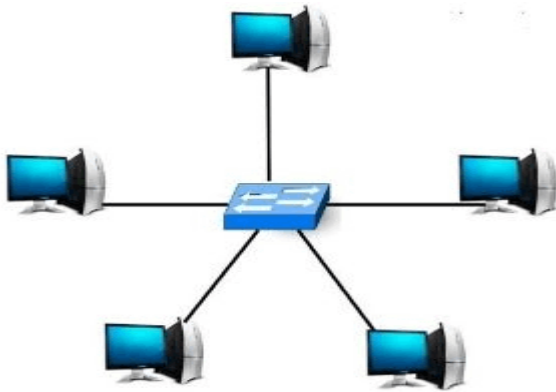
Advantages:

- The data transfer and Communication are easy due to easy packet movement.
- The installation of the network and reconfiguration is easy.
- The presence of errors in links and faulty nodes can be easily identified and isolated.
- The reliability of the ring topology is more.

Disadvantages:

- As the data transmission is unidirectional, the packet will travel all the nodes to reach the destination.
- One node failure incurs damage to the entire network.
- Reconfiguration (adding new nodes) is easy but gets down the performance of the network.
- The delay is more in data transmission when the network contains more stations.

3) Star Topology:



In this topology, all the computers are separately connected to a central node or connection point, which can be a server, a hub, a router, or a switch. This topology offers an advantage that if a cable does not work, only the respective node will suffer, the rest of the nodes will work smoothly. All data or messages that one node sends to another passes through the central hub.

This topology is easy to design and implement as well as it is easy to add additional nodes to the central node. The major drawback of this topology is that it is prone to bottleneck or failure at the central connection point, i.e., failure at the central node will affect the entire communication.

Advantages:

- Easy installation and reconfiguration.
- Less expensive as compared to mesh Topology because the star requires less wiring.
- Easy to identify and isolate the faulty networks and links.
- When we add a new node, there is no interruption to other nodes in the network as the node is directly connected to the hub.
- Using star topology, we can achieve a high speed of data transfers.

Disadvantages

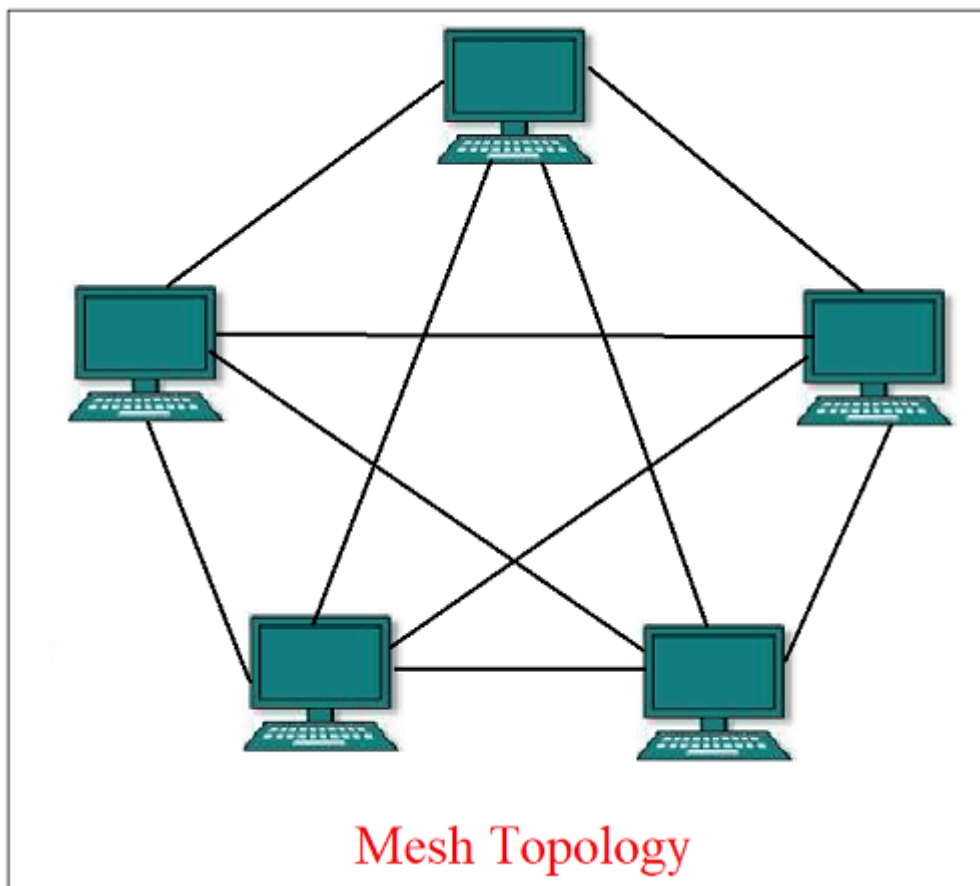
- If the hub fails, the whole network will not work.
- The star topology is more expensive than the bus topology because of the hub.
- We use more wire than that of the bus and ring topology.
- Every node completely depends on the hub for making decisions to transmit and process the data packets.

4) Mesh Topology

In a mesh topology, every device is connected to another device in a network using a point-to-point connection. The connection is generally known as a dedicated connection, as the link transports data between two devices. The number of links in a mesh topology is calculated using the formula below.

Military organizations use mesh topology to avoid breaks down in communications.

Many connections = $n * (n - 1) / 2$. Here, "n" represents the number of nodes in a network.



Advantages:

- Easy to transmit data.
- We can send data from many devices simultaneously. Mesh topology will handle much traffic as compared to other topologies.

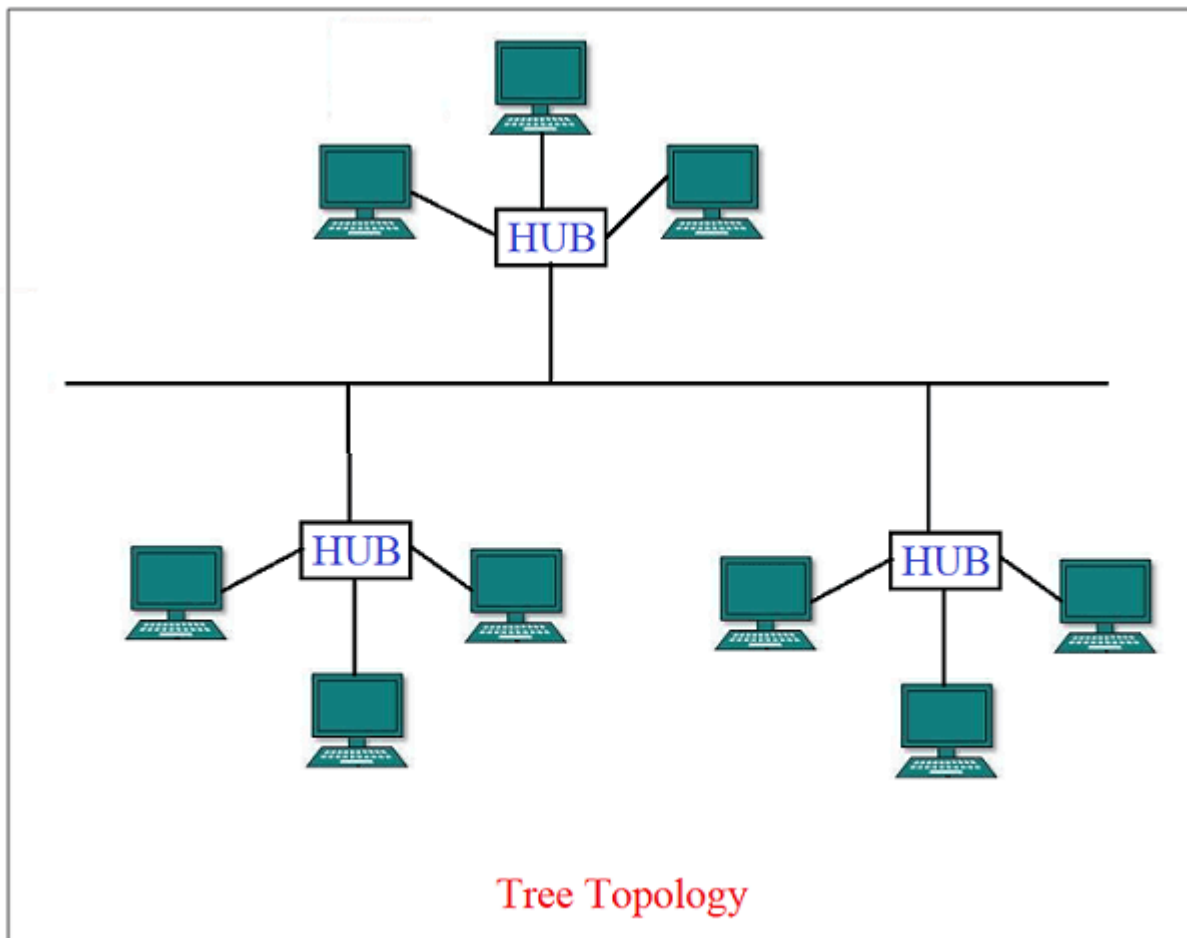
- If one link is broken or remains faulty, data transfer can occur between nodes using other links. Hence data transmission is uninterrupted and reliable.
- The physical margins will not allow other persons to enter and access the messages.
- Fault detection and isolation are easy.

Disadvantages:

- The cabling used to construct the network is more.
- The installation cost is high compared to other topologies due to more wiring.
- The installation and reconfiguration are tough because of the presence of many links.

5) Tree Topology

Tree topology is the combination of star and bus topologies. The nodes are connected to a hub as in star topology, and all the star-connected nodes are placed in a bus topology. The tree topology is a hybrid connection.

**Advantages:**

- Network extension is easy.
- The network is broken into smaller star-connected topology units. The nodes connected in a tree topology are easier to maintain.
- Fault detection and correction are easy.
- If one of the stars connected unit of nodes goes faulty, the other segments will not be affected and network can be runed with remaining nodes.

Disadvantages:

- The main disadvantage is that if the bus cable is damaged, the entire topology will not work.
- The management of Topology becomes easier when the number of nodes and star-connected topologies is increased in large numbers.
- The reconfiguration becomes difficult when new nodes are added.

Q-4) Types of computer network architectures

Computer network architectures are of three types

1) Client-Server architecture

The architecture consists of a system that manages and delivers the services and resources via the network that the client requests. In the client-server architecture, each and every computer or process on the network is considered as a client or server. The servers (like computers) perform various operations to satisfy the client's requests.

The main methodology of client-server architecture is as follows:

- Initially, by using the network-enabled device, the client sends the request.
- Further, the network server acquires, agrees to, and processes the request.
- Lastly, the servers dispatch the reply information to the client.

Examples of client-server architecture include Email servers, Web servers.

2) Peer-to-peer architecture

In the Peer-to-Peer (P2P) architecture, linked computers have the same power and status. There is no central server for processing. Every peer may share some of its resources, like memory and processing power, with the entire computer network. Unlike the client-server architecture, peer-to-peer is a decentralized system in which files will be transferred from one system to another without using a central server.

3) Hybrid Network

The hybrid network architecture includes both client-server architecture and peer-to-peer architecture.

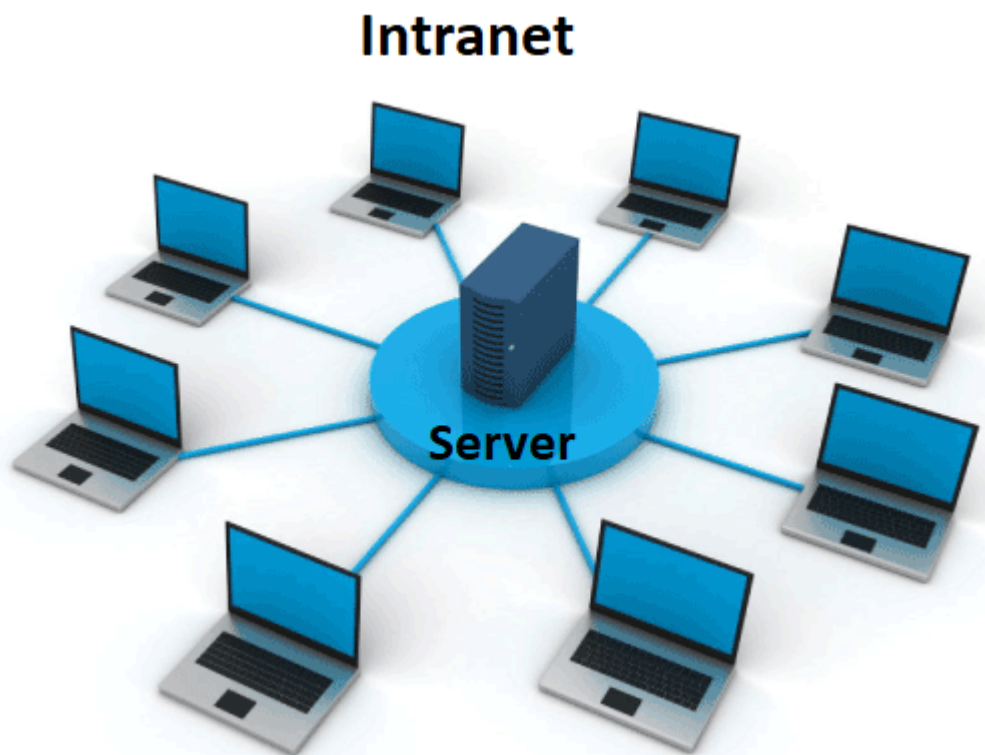
Q-5) What is the Internet?

Internet is a global network that connects billions of computers across the world with each other and to the World Wide Web. It uses standard internet protocol suite (TCP/IP) to connect billions of computer users worldwide. It is set up by using cables such as optical fibers and other wireless and networking technologies. At present, internet is the fastest

mean of sending or exchanging information and data between computers across the world.

It is believed that the internet was developed by "Defense Advanced Research Projects Agency" (DARPA) department of the United States. And, it was first connected in 1969.

Q-6) What is the Intranet?



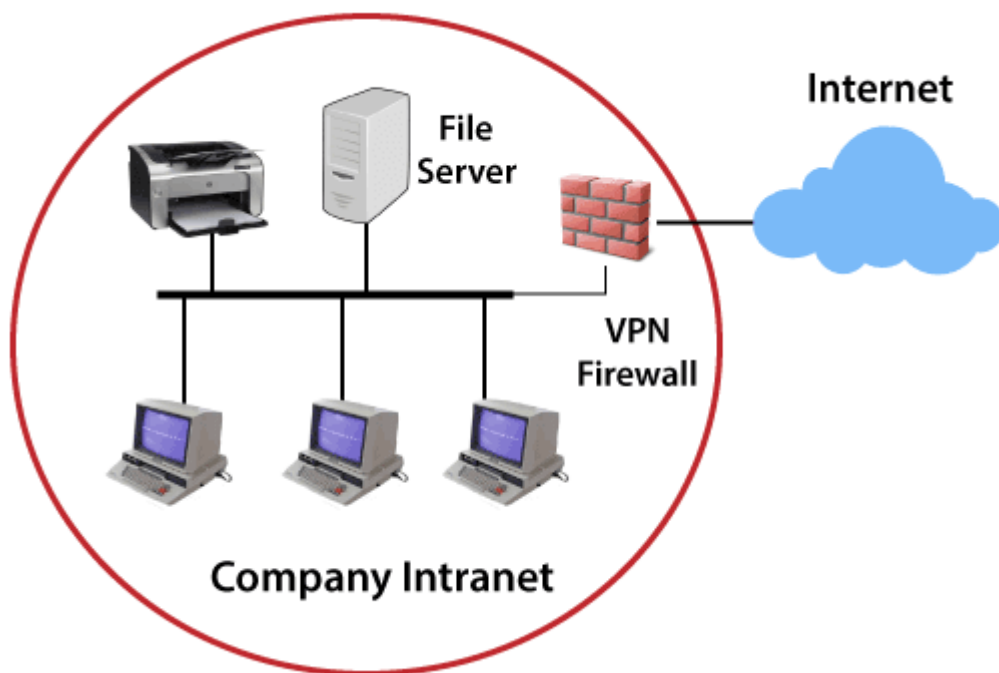
The intranet is a private network that belongs to a particular organization. It is designed for the exclusive use of an organization and its associates, such as employees, customers, and other authorized people. It offers a secure platform to convey information and share data with authorized users. Confidential information, database, links, forms, and applications can be made available to the staff through the intranet. So, it is like a private internet or an internal website that is operating within an organization to provide its employees access to its information and records. Each computer in intranet is identified by a unique IP Address.

It is based on internet protocols (TCP/[IP](#)) and is protected from unauthorized access with firewalls and other security systems. The firewall monitors the incoming and outgoing data packets to ensure they don't

contain unauthorized requests. So, users on the intranet can access the [internet](#), but the internet users can't access the intranet if they are not authorized for it. Furthermore, to access the intranet, the authorized user is required to be connected to its [LAN](#) (Local Area Network).

How the Intranet Works:

Intranet basically comprises three components: a web server, an intranet platform, and applications. The web server is hardware that contains all the intranet software and data. It manages all requests for files hosted over the server and finds the requested files and then delivers it to the user's computer.



The intranet platform, which is software, allows communication tools, collaboration apps, and databases to work seamlessly with each other. It is tailored to the specific needs of a business.

The applications are required to enable users to work smoothly. They are the computing tools that allow users to do their work, communicate, and coordinate with each other and retrieve and store information.

Furthermore, the user who wants to access the intranet is required to have a special network password and should be connected to the LAN. A user who is working remotely can gain access to the intranet through a virtual private network ([VPN](#)) that allows them to sign in to the intranet to access the information.

Q-7) Differentiate between internet and intranet.

Internet	Intranet
It is a medium such as optical fiber cable that connects billions of computers with each other to establish a worldwide network.	It is a small, private network as it belongs to a specific organization.
It has billions of users as it is a public network with a worldwide presence.	It has limited users.
It is not as safe as an intranet.	It is a safer network than the internet.
It can be assessed or used by anyone using an internet-enabled devices, such as laptop, mobile phone, etc.	Only authorized persons can use this network.
It offers a wide range of information, such as news, blogs, websites, etc.	It offers limited information related to its organization's work, policies, updates, etc.
It is not owned by a single person or an organization.	It can be owned by a person or an organization.

Q-8) Explain ISP.

ISP stands for Internet Service Provider. It is a company that provides access to the internet and similar services such as Website designing and virtual hosting. For example, when you connect to the Internet, the connection between your Internet-enabled device and the internet is executed through a specific transmission technology that involves the transfer of information packets through an Internet Protocol route.



Data is transmitted through different technologies, including cable modem, dial-up, DSL, high speed interconnects.

Types of ISP

Accordingly, based on the method of data transmission, the Internet access provided by ISPs can be divided into many types, some of which are as follows:

Dial-up Internet access: It is the oldest technology to provide Internet access by modem to modem connection using telephone lines. In this method, the user's computer is connected to a modem with a telephone line. This method has become outdated today due to slow connection speed. However, in remote areas, this method can be used where the broadband network is not available.

DSL: DSL, which stands for 'digital subscriber line' is an advanced version of the dial-up Internet access method. It uses high frequency to execute a connection over the telephone network and allows the internet and the phone connection to run on the same telephone line.

Wireless Broadband (WiBB): It is a modern broadband technology for Internet access. It allows high-speed wireless internet within a large area. To use this technology, you are required to place a dish on the top of your house and point it to the transmitter of your Wireless Internet Service Provider (WISP).

Wi-Fi Internet: It is the short form for "wireless fidelity," which is a wireless networking technology that provides wireless high-speed Internet connections using radio waves. To use the internet, you are required to be

within the range of wi-fi network. It is commonly used in public places such as hotels, airports, restaurants to provide internet access to customers.

ISDN: It is a short form of Integrated Services Digital Network. It is a telephone system network which integrates a high-quality digital transmission of voice and data over the same standard phone line.

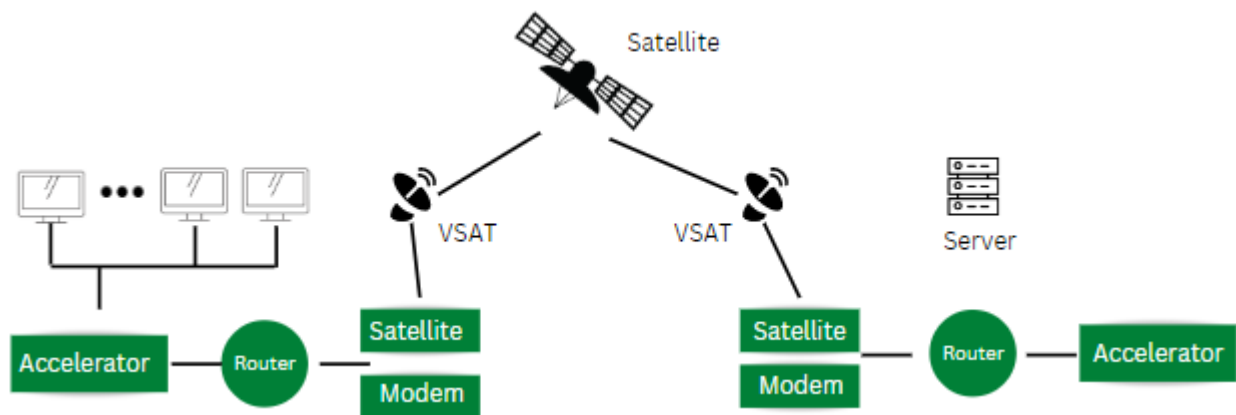
Ethernet: It is a wired LAN (Local Area Network) where computers are connected within a primary physical space. It enables devices to communicate with each other via a protocol (a set of rules or common network language). It may provide different speeds such as 10 Mbps, 100 Mbps and 10 Gbps.

List of ISP

- Reliance Jio
- Vodafone Idea
- Airtel
- BSNL
- Hathway

Q-9) Explain VSAT.

VSAT stands for **Very Small Aperture Terminal**. VSAT is of importance, especially in remote areas such as the hilly mountain regions, where Internet connectivity cannot be directly provided. There is no option left in such places through which the Internet can be accessed anyway. These places could be the Ocean region and the Sea region, where the citizens have a lack of utilities. In such cases, VSAT is one of the technologies that make Internet Connectivity possible so that people can access it. All the work is carried out in a workstation group where the sending and receiving of signals takes place. The data such as audio, video, etc, can be sent and received. Therefore, VSAT is considered to be a private Earth Station. This Earth Station is designed in a way such that the signals are transmitted and received with the help of satellites. The term 'Very Small' in VSAT (Very Small Aperture Terminal) refers to the antenna size that is very small.



Characteristics of VSAT:

- VSAT is a two-way communication earth satellite station that has an antenna of less than 3 meters.
- The size of the VSAT antenna may vary from 75 cm to 1.2 meters.
- The data ranges between 4 kilobytes per second to 16 megabytes per second in VSAT.

Working for VSAT:

- The end user of the set requires a box that acts as an interface between the user and its system.
- The box consists of an antenna and a transceiver.
- The work of this transceiver is to send and receive signals to and from the transponder located in space.
- Every workstation works like a hub. The satellite sends or receives a signal from it.
- Each user is connected to each of the interconnected hub stations and operates in the form of a Star Topology (There is a server in between and all the nodes are connected to it, forming a star-like figure).
- With this medium, we can transmit data, voice, and video signals.

Advantages of VSAT:

- VSAT Terminals and its Hardware can be installed in vehicles such as Trucks or Vans and can also be used in situations where mobility is a need.
- Audio, Video, and data signals can be transmitted and received efficiently.

- Internet Access: A VSAT Network also serves to provide internet access in addition to pointing to a WAN link.
- It is set in the consumer broadband industry to make it the leading major wave. VSAT network provides an “Always on” broadband Internet Service.
- Information from remote locations can be accessed using satellites.
- It is more used for connectivity in rural areas, ships, and coastal regions.
- Mobile access is another conventional strength of satellite networks. For example, we can do online surfing, watch TV, use applications, and much more.
- VSAT Networks are not affected by earthquakes, Cyclones, and other natural calamities.
- VSAT Networks, with a low-cost architecture and extra powerful systems, share digital information.

Disadvantages of VSAT:

- **Delay:** VSAT Technology uses satellites in Geosynchronous orbit. This type of data transmission has an approximate delay of about 500 milliseconds for each round trip. It introduces a problem with the application that requires a consistent transmission.
- **Environmental conditions:** Like other satellite systems, the VSAT network may also get affected by the weather and other environmental conditions. The signal strength may be weak at times, although it depends upon the size of the antenna, frequency band, and the power of the transmitters.
- **Clear view:** Since VSAT requires an external antenna, therefore to contact the satellite, the location must have a clear view of the southern sky.

Q-10) Explain URL.

URL stands for Uniform Resource Locator. It is the address of a resource, which can be a specific webpage or a file, on the internet. It is also *known as web address* when it is used with http. It was created in 1994 by Tim Berners-Lee. URL is a specific character string that is used to access data from the World Wide Web. It is a type of URI (Uniform Resource Identifier).

Every URL contains the following information:

- The scheme name or protocol.
- A colon, two slashes.
- A host, normally called a domain name but sometimes as a literal IP address.
- A colon followed by a port number.
- Full path of the resource.

The URL of a web page is displayed above on the page in the address bar. A typical URL looks like this:

`http://www.javatpoint.com/full-form`

The above URL contains:

- **protocol:** http
- **host or domain:** www.javatpoint.com
- **Path of the resource:** /full-form

Q-11) Explain URI.

URI stands for Uniform Resource Identifier. It is a generic term for all the name and addresses which show objects on the World Wide Web. It is generally a sequence of characters which identifies a logical resource or the name and location of a file or resource in a uniform format.

A URI can be of two types: Uniform Resource Locator (URL) and Uniform Resource Names (URNs). It enables resources to be accessed by other computers across a network or over the World Wide Web.

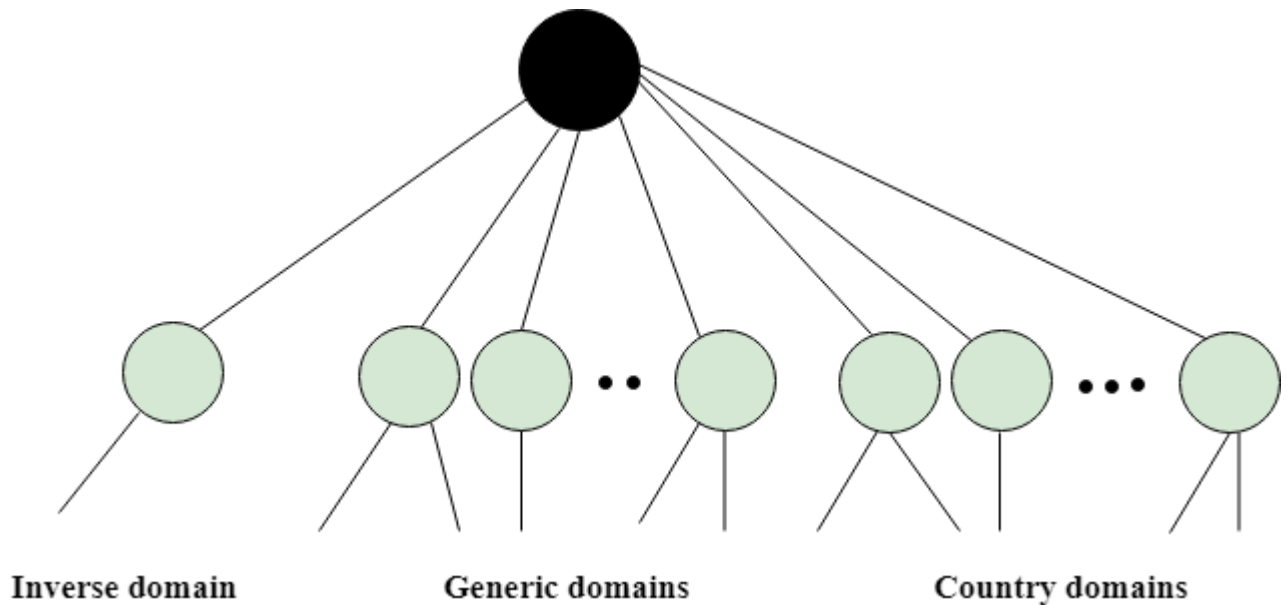
Q-12) Explain DNS.

An application layer protocol defines how the application processes running on different systems, pass the messages to each other.

- DNS stands for Domain Name System.
- DNS is a directory service that provides a mapping between the name of a host on the network and its numerical address.
- DNS is required for the functioning of the internet.
- Each node in a tree has a domain name, and a full domain name is a sequence of symbols specified by dots.
- DNS is a service that translates the domain name into IP addresses. This allows the users of networks to utilize user-friendly names when looking for other hosts instead of remembering the IP addresses.

- For example, suppose the FTP site at EduSoft had an IP address of 132.147.165.50, most people would reach this site by specifying ftp.EduSoft.com. Therefore, the domain name is more reliable than IP address.

DNS is a TCP/IP protocol used on different platforms. The domain name space is divided into three different sections: generic domains, country domains, and inverse domain.



Generic Domains

- It defines the registered hosts according to their generic behavior.
- Each node in a tree defines the domain name, which is an index to the DNS database.
- It uses three-character labels, and these labels describe the organization type.

Label	Description
Aero	Airlines and aerospace companies
Biz	Businesses or firms
Com	Commercial Organizations
Coop	Cooperative business Organizations

Edu	Educational institutions
Gov	Government institutions
Info	Information service providers
Int	International Organizations
Mil	Military groups
museum	Museum & other nonprofit organizations
Name	Personal names
Net	Network Support centers
Org	Nonprofit Organizations
Pro	Professional individual Organizations

Country Domain

The format of country domain is same as a generic domain, but it uses two-character country abbreviations (e.g., us for the United States) in place of three character organizational abbreviations.

Inverse Domain

The inverse domain is used for mapping an address to a name. When the server has received a request from the client, and the server contains the files of only authorized clients. To determine whether the client is on the authorized list or not, it sends a query to the DNS server and ask for mapping an address to the name.

Q-13) Explain WWW.

World Wide Web, which is also known as a Web, is a collection of websites or web pages stored in web servers and connected to local computers through the internet. These websites contain text pages, digital images, audios, videos, etc. Users can access the content of these sites from any part of the world over the internet using their devices such as computers, laptops, cell phones, etc. The WWW, along with internet, enables the retrieval and display of text and media to your device.

The building blocks of the Web are web pages which are formatted in HTML and connected by links called "hypertext" or hyperlinks and accessed by HTTP. These links are electronic connections that link related pieces of information so that users can access the desired information quickly. Hypertext offers the advantage to select a word or phrase from text and thus to access other pages that provide additional information related to that word or phrase.

A web page is given an online address called a Uniform Resource Locator (URL). A particular collection of web pages that belong to a specific URL is called a website, e.g., *www.facebook.com*, *www.google.com*, etc. So, the World Wide Web is like a huge electronic book whose pages are stored on multiple servers across the world.

The World Wide Web was invented by a British scientist, Tim Berners-Lee in 1989. He was working at CERN at that time.

Q-14) Explain Search Engine.

A search engine is an **online answering machine**.

Search Engine refers to a huge database of internet resources such as web pages, newsgroups, programs, images etc. It helps to locate information on world wide web.

User can search for any information by passing query in form of keywords or phrase. Then it searches for relevant information in its database and return to the user. The process of organizing content in the form of a list is commonly known as a **Search Engine Results Page (SERP)**.

Google, Yahoo!, Bing, YouTube are some popular examples of search engines.

Components of Search Engine

There are the following four basic components of Search Engine -

1. Web Crawler

Web Crawler is also known as a **search engine bot, web robot, or web spider**. It plays an essential role in search engine optimization (SEO) strategy. It is mainly a software component that traverses on the web, then downloads and collects all the information over the Internet.

2. Database

The search engine database is a type of **Non-relational database**. It is the place where all the web information is stored. It has a large number of web resources. Some most popular search engine databases are **Amazon Elastic Search Service** and **Splunk**.

3. Search Interfaces

Search Interface is one of the most important components of Search Engine. It is an interface between the user and the database. It basically helps users to search for queries using the database.

Q-15) What is RemoteLogin?

Remote Login is a process in which user can login into remote site i.e. computer and use services that are available on the remote computer. With the help of remote login a user is able to understand result of transferring and result of processing from the remote computer to the local computer.

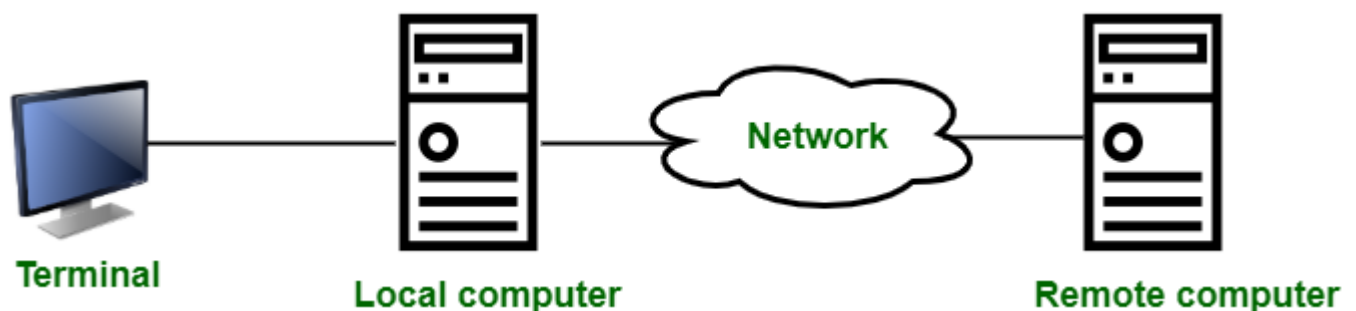


Figure – Remote login

It is implemented using [Telnet](#).

Q-16) what is TelNet?

TELNET stands for **Teletype Network**. It is a type of protocol that enables one computer to connect to the local computer. It is used as a standard [TCP/IP protocol](#) for virtual terminal service which is provided by [ISO](#). The computer which starts the connection is known as the **local computer**.

The computer which is being connected to i.e. which accepts the connection known as the **remote computer**.

During telnet operation, whatever is being performed on the remote computer will be displayed by the local computer. Telnet operates on a

client/server principle. The local computer uses a telnet client program and the remote computers use a telnet server program.

Logging

The logging process can be further categorized into two parts:

1. Local Login
2. Remote Login

1. Local Login: Whenever a user logs into its local system, it is known as local login.

The Procedure of Local Login

- Keystrokes are accepted by the terminal driver when the user types at the terminal.
- Terminal Driver passes these characters to OS.
- Now, OS validates the combination of characters and opens the required application.

2. Remote Login: [Remote Login](#) is a process in which users can log in to a remote site i.e. computer and use services that are available on the remote computer. With the help of remote login, a user is able to understand the result of transferring the result of processing from the remote computer to the local computer.

The Procedure of Remote Login

- When the user types something on the local computer, the local operating system accepts the character.
- The local computer does not interpret the characters, it will send them to the TELNET client.
- TELNET client transforms these characters to a universal character set called [Network Virtual Terminal \(NVT\)](#) characters and it will pass them to the local TCP/IP protocol Stack.
- Commands or text which are in the form of NVT, travel through the Internet and it will arrive at the [TCP/IP](#) stack at the remote computer.
- Characters are then delivered to the operating system and later on passed to the TELNET server.
- Then TELNET server changes those characters to characters that can be understandable by a remote computer.

- The remote operating system receives characters from a pseudo-terminal driver, which is a piece of software that pretends that characters are coming from a terminal.
- The operating system then passes the character to the appropriate application program.

Q-17) What is E-mail?

Email stands for [Electronic Mail](#). It is a method to send messages from one computer to another computer through the internet. It is mostly used in business, education, technical communication, document interactions. It allows communicating with people all over the world without bothering them. In 1971, a test email sent Ray Tomlinson to himself containing text.

It is the information sent electronically between two or more people over a network. It involves a sender and receiver/s.

Email messages include three components, which are as follows:

- **Message envelope:** It depicts the email's electronic format.
- **Message header:** It contains email subject line and sender/recipient information.
- **Message body:** It comprises images, text, and other file attachments.

The email was developed to support rich text with custom formatting, and the original email standard is only capable of supporting plain text messages. In modern times, email supports HTML (Hypertext markup language), which makes it capable of emails to support the same formatting as websites. The email that supports HTML can contain links, images, CSS layouts, and also can send files or "email attachments" along with messages.

Q-18) Explain E-Commerce.

E-commerce stands for electronic commerce.

E-commerce is the buying and selling of goods and services, or the transmitting of funds or data, over an electronic network, primarily the internet. These business transactions occur either as business-to-business

([B2B](#)), business-to-consumer ([B2C](#)), consumer-to-consumer or consumer-to-business.

Types of E-Commerce

Types of E-commerce model:

There are [types of e-commerce models](#) that can describe almost every transaction that takes place between consumers and businesses.

1. **Business to Consumer (B2C):** When a good or service is sold to an individual consumer by a business, e.g., we buy a pair of shoes from an online retailer.
2. **Business to Business (B2B):** When a good or service is sold by a business to another business, e.g., a software-as-a-service is sold by a business for other businesses to use.
3. **Consumer to Consumer (C2C):** When a good or service is sold by a consumer to another consumer, e.g., we sell our old furniture on eBay to another consumer.
4. **Consumer to Business (C2B):** When a consumer's own products or services is sold to a business or organization, e.g., an authority offers exposure to their online audience in exchange for a fee or a photographer licenses their photo for a business to use.
5. **Business-to-government (B2G):** In this model, businesses sell products or services to government agencies through online portals or marketplaces.
6. **Government-to-business (G2B):** In this model, government agencies sell products or services to businesses through online portals or marketplaces.
7. **Government-to-consumer (G2C):** In this model, government agencies provide products or services directly to consumers through online portals or marketplaces.

Business Application of E-commerce:

- Conversational commerce: e-commerce via chat
- Digital Wallet
- Newsgroups
- Online banking

Advantages of E-commerce:

- E-commerce enables fast and secure shopping.
- It is making digitalized world.
- E-commerce also enables to choose different goods and services according to your choice.
- It is a simple way of selling and buying products and services.
- E-commerce replaced the paper work as all transactions are through internet today.
- It provides better management system, as it has a centralized database.
- E-commerce via internet covers a large number of customers worldwide.
- E-commerce has several payment modes.

Disadvantages of E-commerce:

- E-commerce has no universal standard for quality and reliability.
- E-commerce works through internet, it is possible that navigation on internet itself may be slow.
- Strong security is required in e-commerce as all transactions are through internet.
- There is high risk of buying unsatisfactory products through e-commerce.
- It uses public key infrastructure which is not safe.
- Customers also trap in banking fraud which is quite frequent.
- Hackers also try to get access of data or to destroy data in e-commerce.

Q-19) What is E-Business?

E-business, commonly known as electronic or online business is a business where an online transaction takes place. In this transaction process, the buyer and the seller do not engage personally, but the sale happens through the internet. In 1996, Intel's marketing and internet team coined the term "E-business".

Features of E-Business

Here are the few features of e-business:

- Easy setup
- No geographic barriers
- Cost-effective

- Flexible trading hours
- Cheap marketing policy
- No interaction between buyer and seller
- Delivery of goods takes extra time
- Transaction threat is prominent than traditional business
- People can buy any goods and services from anywhere and at anytime

Q-20) Explain E-Governance.

Electronic Governance or E-Governance is the application of Information and Communication Technology (ICT) for providing government services, interchange of statistics, communication proceedings, and integration of various independent systems and services. Through the means of e-governance, government services are made available to citizens in a suitable, systematic, and transparent mode. The three main selected groups that can be discriminated against in governance concepts are government, common people, and business groups.

Elements of E-Governance:

Basic elements of e-governance are:

1. Government
2. Citizens
3. Investors/Businesses

Types of E-Governance:

There are four different ways to interact with the government:

1. **G2G- (Government to Government)** is a concept that strives to facilitate information sharing between Governments. Examples include the exchange of government documents and information between the police departments of various States.
2. **G2C (Government to Citizen)** is a concept that strives to facilitate information exchange between the government and its constituents. Examples include online complaint filing, online water and energy bill paying, and more.
3. **G2B (Government to Business)** is a concept that tries to facilitate information exchange between the public and commercial sectors.

Examples include sharing laws and data, tax collection, and company patent clearance.

4. **G2E (Government to Employees)** is a concept that intends to share information between the government and its workers, allowing workers to complete various forms online.

Advantages of E-Governance:

The supreme goal of e-governance is to be able to provide an increased portfolio of public services to citizens in a systematic and cost effective way. It allows for government transparency because it allows the public to be informed about what the government is working on as well as the policies they are trying to implement.

The main advantage while executing electronic government will be to enhance the efficiency of the current system.

Another advantage is that it increases transparency in the administration, reduces costs, increases revenue growth, and also improves relationships between the public and the civic authorities.

Disadvantages of E-Governance:

The main disadvantage regarding e-governance is the absence of fairness in public access to the internet, of trustworthy information on the web, and disguised agendas of government groups that could have an impact and could bias public opinions.

Q-21) Explain M-Commerce.

M-commerce, or mobile commerce, is a subset of e-commerce that refers specifically to transactions conducted using mobile devices, such as smartphones and tablets. M-commerce includes not only online purchases made through mobile devices, but also mobile payments, mobile banking, and other transactions facilitated by mobile technology.

Types of m-commerce

M-commerce is categorized based on the following three basic functions:

- **Mobile shopping** enables customers to buy a product using a mobile device with an application such as [Amazon](#) or a web app. A

subcategory of mobile shopping is app commerce, which is a transaction that takes place over a [native app](#).

- **Mobile banking** is online banking designed for handheld technology. It enables customers to access accounts and brokerage services, conduct financial transactions, pay bills and make stock trades. This is typically done through a secure, dedicated app provided by the banking institution.
- **Mobile payments** are an alternative to traditional payment methods, such as cash, check, credit and debit cards. They enable users to buy products in person using a mobile device. Digital wallets, such as Apple Pay, let customers buy products without swiping a card or paying with cash. Mobile payment apps, such as PayPal.

Q-22) Difference between E-commerce and M-commerce.

S.No.	E-commerce	M-commerce
01.	Electronic Commerce in short it is called as e-commerce.	Mobile Commerce in short it is called as m-commerce.
02.	In general, e-commerce activities are performed with the help of desktop computers and laptops.	M-commerce activities are performed with the help of mobile devices like smartphones, tablets, PDA's (Personal Digital Assistant) etc.
03.	E-commerce is an older concept.	M-commerce is an newer concept.
04.	It is broad term which refers doing shopping and making payments online with help of electronic devices like Laptop and computers.	It is subcategory of ecommerce which does the same this via mobile devices.
05.	In e-commerce the use of internet is mandatory	But in case of m-commerce some activities can be performed without internet also.
06.	E-commerce devices are not easy to carry and portability point of view it is not so good.	M-commerce devices are easy to carry and portability point of view it is good.
07.	E-commerce developed in 1970's.	M-commerce developed in 1990's.

08.	Its reachability is comparatively low than the m-commerce as it is not so good in portability.	Its reachability is more than that of e-commerce only due to the use of mobile devices.
09.	In e-commerce location tracking capabilities are limited due to the non-portability of devices.	In m-commerce location tracking capabilities is so good as mobile apps track and identify user locations with the help of GPS technology, Wi-Fi, and so on.
10.	E-commerce fails in push notification.	In m-commerce push notification can be achieved.
11.	E-commerce is conducted using desktop or laptop computers.	M-commerce is conducted using mobile devices such as smartphones and tablets.
12.	E-commerce typically requires a stable internet connection and a computer.	M-commerce allows consumers to shop and make purchases from anywhere
13.	E-commerce transactions typically rely on credit cards and other traditional payment methods.	M-commerce offers a wider range of payment options, including mobile wallets and contactless payments.
14.	Examples of E-commerce includes Amazon, Flipkart, Quikr, Olx websites.	Examples of M-commerce includes mobile banking like paytm, in-app purchasing Amazon mobile app.

Q-23) What is webpage?

A **webpage** is a digital document that is linked to the World Wide Web and viewable by anyone connected to the internet having a web browser. It can contain any type of information, such as text, color, graphics, animations, videos, sounds, etc.

A **webpage** is a document that is written in the [HTML](#), it can be viewed from the Internet. It can be accessed by entering the URL on the address bar of the web browser.

Q-24) What is webbrowser?

A web browser or simply browser is application software used to access the internet. Some examples of Web browsers are **Google Chrome, Microsoft Internet Explorer, Safari, etc.** It does two things:

- It connects to a web server on the internet and requests a page that the user wants to view; once it finds that page, it displays it on its device.
- It can interpret the set of HTML tags within a page to display the page in the correct format.

Q-25) What is webserver?

A web server can be understood as a computer that hosts or provide a website on the internet. It contains webserver software and component files of a website such as *HTML document, images, CSS stylesheet, and JS files*.

Example of web server software are apache,iis.

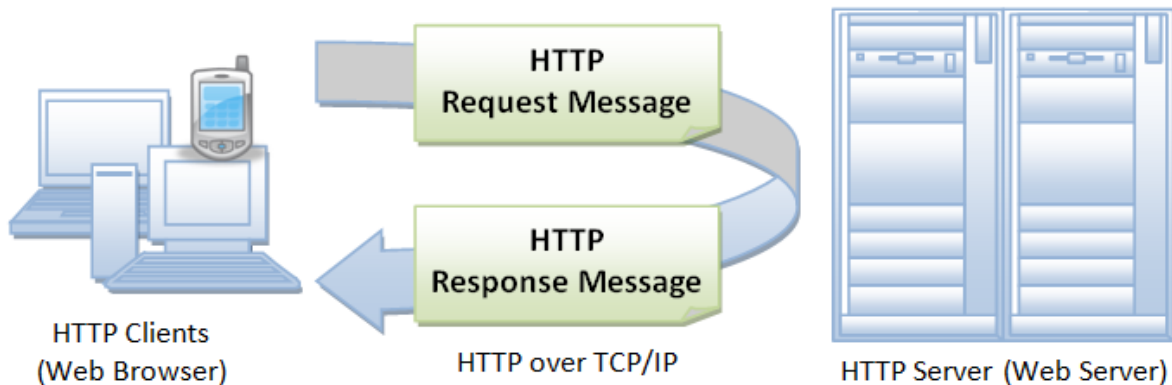
Q-26) What is webhosting?

Web hosting is a service that stores your website or web application and makes it easily accessible across different devices such as desktop, mobile, and tablets. Any web application or website is typically made of many files, such as images, videos, text, and code, that you need to store on special computers called servers. The web hosting service provider maintains, configures, and runs physical servers that you can rent for your files. Web site and web application hosting services also provide additional support, such as security, website backup, and website performance, which free up your time so that you can focus on the core functions of your website.

Q-27) Explain HTTP.

- HTTP stands for **HyperText Transfer Protocol**.
- It is a protocol used to access the data on the World Wide Web (www).
- The HTTP protocol can be used to transfer the data in the form of plain text, hypertext, audio, video, and so on.
- This protocol is known as HyperText Transfer Protocol because of its efficiency that allows us to use in a hypertext environment where there are rapid jumps from one document to another document.
- HTTP is similar to the FTP as it also transfers the files from one host to another host. But, HTTP is simpler than FTP as HTTP uses only one connection, i.e., no control connection to transfer the files.

- HTTP is used to carry the data in the form of MIME-like format.
- HTTP is similar to SMTP as the data is transferred between client and server. The HTTP differs from the SMTP in the way the messages are sent from the client to the server and from server to the client. SMTP messages are stored and forwarded while HTTP messages are delivered immediately.



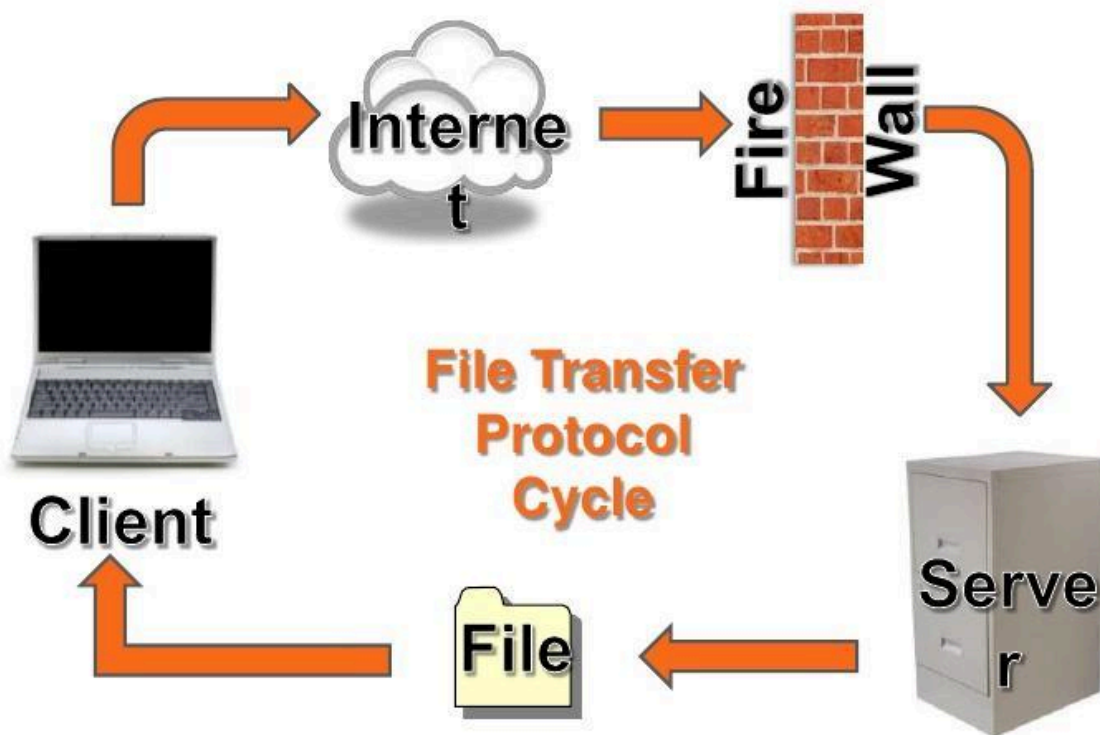
The above figure shows the HTTP transaction between client and server. The client initiates a transaction by sending a request message to the server. The server replies to the request message by sending a response message.

Features of HTTP:

- **Connectionless protocol:** HTTP is a connectionless protocol. HTTP client initiates a request and waits for a response from the server. When the server receives the request, the server processes the request and sends back the response to the HTTP client after which the client disconnects the connection. The connection between client and server exist only during the current request and response time only.
- **Media independent:** HTTP protocol is a media independent as data can be sent as long as both the client and server know how to handle the data content. It is required for both the client and server to specify the content type in MIME-type header.
- **Stateless:** HTTP is a stateless protocol as both the client and server know each other only during the current request. Due to this nature of the protocol, both the client and server do not retain the information between various requests of the web pages.

Q-27) Explain FTP.

FTP stands for File Transfer Protocol. FTP is a standard network protocol used to transfer files between computers (a client and server) over a TCP/IP network. It is a function of Application layer and built on client-server architecture. Client controls the conversation, while server transmits the file content. Browser acts as a client and starts the conversation by making some request on the server. Through FTP, a client can remove, download, delete, or upload files on a server.

**Objectives of FTP**

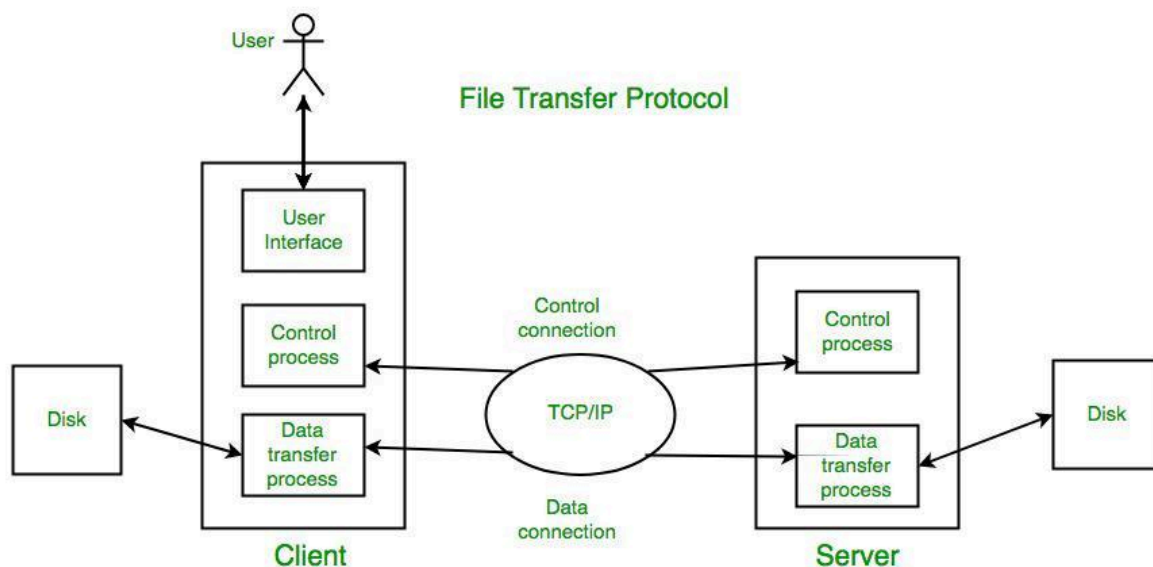
- It provides the sharing of files.
- It is used to encourage the use of remote computers.
- It transfers the data more reliably and efficiently.

Why FTP?

Although transferring files from one system to another is very simple and straightforward, but sometimes it can cause problems. For example, two

systems may have different file conventions. Two systems may have different ways to represent text and data. Two systems may have different directory structures. FTP protocol overcomes these problems by establishing two connections between hosts. One connection is used for data transfer, and another connection is used for the control connection.

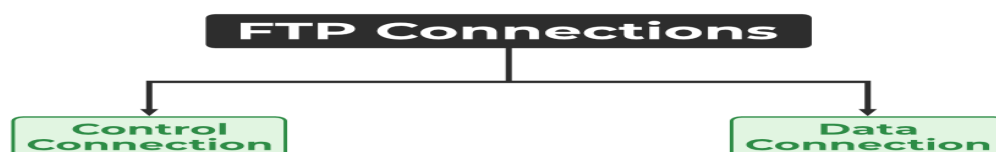
Mechanism of FTP



The above figure shows the basic model of the FTP. The FTP client has three components: the user interface, control process, and data transfer process. The server has two components: the server control process and the server data transfer process.

Types of Connection in FTP

1. Control Connection
2. Data Connection



- **Control Connection:** The control connection uses very simple rules for communication. Through control connection, we can transfer a line of command or line of response at a time. The control connection

is made between the control processes. The control connection remains connected during the entire interactive FTP session.

- **Data Connection:** The Data Connection uses very complex rules as data types may vary. The data connection is made between data transfer processes. The data connection opens when a command comes for transferring the files and closes when the file is transferred.

FTP Clients

- FTP client is a program that implements a file transfer protocol which allows you to transfer files between two hosts on the internet.
- It allows a user to connect to a remote host and upload or download the files.
- It has a set of commands that we can use to connect to a host, transfer the files between you and your host and close the connection.

Advantages of FTP

- Speed is one of the advantages of FTP (File Transfer Protocol).
- File sharing also comes in the category of advantages of FTP in this between two machines files can be shared on the network.
- Efficiency is more in FTP.

Disadvantages of FTP

- File size limit is the drawback of FTP only 2 GB size files can be transferred.
- Multiple receivers are not supported by the FTP.
- FTP does not encrypt the data this is one of the biggest drawbacks of FTP.
- FTP is unsecured we use login IDs and passwords making it secure but they can be attacked by hackers.

Q-28) What is website?

A collection of web pages and media over the internet is called a website. The website can be published on at least one web server and can be identified by a common domain name, for example - javatpoint.com, google.com, youtube.com.

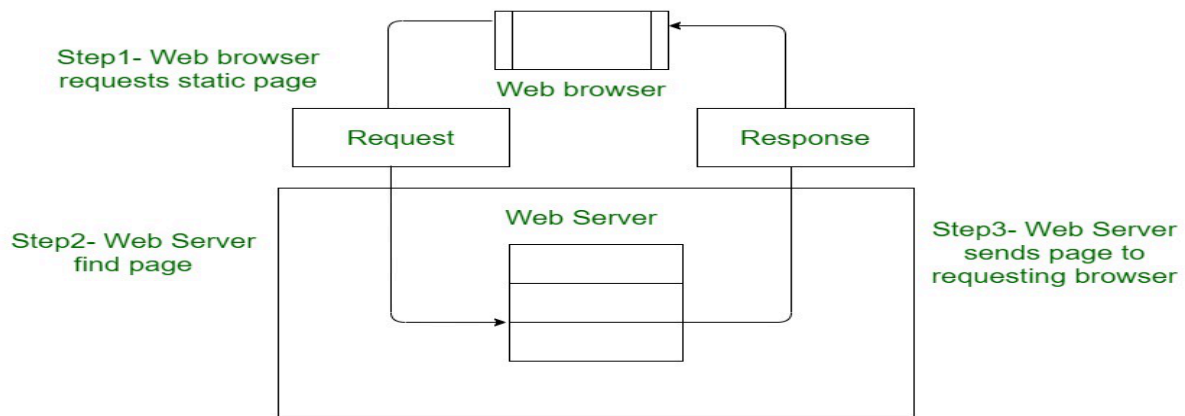
Types of Website

- Static Website
- Dynamic Website

Q-29) Explain Static Website.

Static website is the basic type of website that is easy to create. You don't need the knowledge of web programming and database design to create a static website. Its web pages are coded in HTML.

The codes are fixed for each page so the information contained in the page does not change and it looks like a printed page.



Q-29) Explain Dynamic Website.

Dynamic website is a collection of dynamic web pages whose content changes dynamically. It accesses content from a database or Content Management System (CMS). Therefore, when you alter or update the content of the database, the content of the website is also altered or updated.

Dynamic website uses client-side scripting or server-side scripting, or both to generate dynamic content.

Client side scripting generates content at the client computer on the basis of user input. The web browser downloads the web page from the server and processes the code within the page to render information to the user.

In server side scripting, the software runs on the server and processing is completed in the server then plain pages are sent to the user.



Q-30) Difference Between Static and Dynamic Websites:

Static Website	Dynamic Website
Content of Web pages can not be change at runtime.	Content of Web pages can be changed.
No interaction with database possible.	Interaction with database is possible
It is faster to load as compared to dynamic website.	It is slower than static website.
Cheaper Development costs.	More Development costs.
No feature of Content Management.	Feature of Content Management System.
HTML, CSS, Javascript is used for developing the website.	Server side languages such as PHP, ASP.Net, SERVLET are used.
Same content is delivered everytime the page is loaded.	Content may change everytime the page is loaded.

Q-31) Explain Responsive Website.

Responsive Web Design comprises two words i.e., responsive and web design. Responsive means to respond and web design means to design website. Therefore, responsive web design generally means website that respond to or resize or adjust itself depending upon screen size it is being seen through. It automatically adjusts to fit user's screen whether it's desktop, laptop, mobile, tablet, etc. It only uses one layout for web page and it can be done either using CSS and HTML or CSS3 and HTML5.